FLOWCHART TEKNIS — HOVERBOARD-VESC

rofiqcp/hoverboard-vesc • branch v1 • STM32F103RCT6 • VESC 6.00 dual-motor FOC

I. INISIALISASI

- **START**
- main() → HAL_Init(); aktifkan AFIO
- **SystemClock_Config() gagal?** — TIDAK / YA
- MX_GPIO_Init(); MX_TIM_Init()
- MX_ADC1_Init(); MX_ADC2_Init(); BLDC_Init()
- Input_Lim_Init(); Input_Init()
- Start ADC1/ADC2; __enable_irq() setelah periferal siap

II. MAIN LOOP / TASK

- **while(1)**
- mcpwm_foc_outer_control_non_isr(vesc_now_ms)
- platform_watchdog_service()
- usart3_rx_check() — drain USART3 DMA
- vesc_protocol_process_pending()
- Update status / telemetri VESC; ulangi

III. ISR / KONTROL

- **ADC dual-DMA + kontrol FOC interrupt-driven**
- TIM1 / TIM8 advanced PWM timers
- Regulasi arus motor kiri & kanan
- Iq_target + estimasi Hall / rotor
- Update TIM1->CCR1..3 dan TIM8->CCR1..3
- USART3 PB10/PB11: VESC, 115200 baud

IV. FAULT & SAFETY

- startup_fail_safe(stage): disable IRQ
- TIM1/TIM8 BDTR.MOE=0; CCR1..3=0
- **controllerFaultActive(): motor fault?** — TIDAK / YA
- Fault aktif → release / force bridges off
- Watchdog & USART3 DMA health monitoring
- Fault/status dikirim via protokol bila tersedia

DETAIL SOURCE CODE / CATATAN

Identifier terverifikasi di Src/main.c: main(), SystemClock_Config(), MX_GPIO_Init(), MX_TIM_Init(), MX_ADC1_Init(), MX_ADC2_Init(), BLDC_Init(), Input_Lim_Init(), Input_Init(), mcpwm_foc_outer_control_non_isr(), usart3_rx_check(), vesc_protocol_process_pending(), controllerFaultActive(), startup_fail_safe(). SysTick_IRQn priority (3,0); exception priorities (0,0). FOC konsep: Clarke abc→αβ; Park αβ→dq; regulator arus; inverse transform/PWM. Nama ISR khusus perlu dilihat pada modul motor.